

01 / Newton's laws

Newton's first law: A body remains at rest or moves in a straight line at constant velocity unless a net external force acts on it.

Newton's second law: Net force equals mass times acceleration: $F = m a$. Force is measured in newtons (N), mass in kilograms (kg), and acceleration in metres per second squared (m/s^2).

Worked example: A 2 kg body accelerating at 4 m/s^2 experiences a net force of 8 N.

Newton's third law: Forces in an interaction are equal in magnitude and opposite in direction. They act on different bodies, so they do not cancel as forces on one body.

02 / Energy and momentum

Kinetic energy is the energy of motion. For a body of mass m moving at speed v , kinetic energy is $E = (1/2) m v^2$. Energy is measured in joules (J).

Worked example: A 2 kg body moving at 3 m/s has kinetic energy of 9 J.

Linear momentum equals mass times velocity: $p = m v$. Its SI unit is kg m/s.

Worked example: A 2 kg body moving at 3 m/s has momentum of 6 kg m/s.

03 / Work and power

When a constant force acts parallel to displacement, work equals force times displacement: $W = F d$. Work is measured in joules (J).

Worked example: A 10 N force acting along a displacement of 3 m does 30 J of work.

Average power is work divided by elapsed time: $P = W/t$. Power is measured in watts (W).

Worked example: Doing 30 J of work in 6 seconds gives an average power of 5 W.